THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 11 & 12: **Areas Related to Circles, Surface Areas and Volumes** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $\frac{77}{3}$ cm²	$\frac{1}{6} \times 154$
2	**(a)** 44 cm	$\frac{1}{3} \times 132$
3	**(a)** 38.5 cm²	A quadrant
4	**(c)** Incorrect — it is πr	Half of $2\pi r$
5	**(b)** Student A is correct	$\frac{360^\circ}{4}$
6	**(c)** 18π cm³	$\frac{2}{3}\pi \times 27$
7	**(b)** 440 cm²	$2 \times \frac{22}{7} \times 7 \times 10$
8	**(a)** $4\pi r^2$	It is a sphere
9	**(b)** Incorrect — it is $\pi r l + 2\pi r^2$	No flat face is exposed
10	**(a)** Student A is correct	B forgot the $\frac{1}{3}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{1}{12} \times 3.14 \times 144 = 37.68$ cm²	1+1
12	$\frac{1}{4} \times 88 = 22$ cm	2
13	$462 + 154 = 616$ cm³	1+1
14	$20 \times 10 \times 10$: $2(200 + 100 + 200) = 1000$ cm²	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sector ≈ 209.33; equilateral triangle $= \frac{\sqrt{3}}{4} \times 400 = 173$ ⇒ segment ≈ 36.33 cm²	1+1+1
16	$l = 10$: cone 60π, cylinder CSA 120π, base 36π ⇒ $216\pi \approx 678.86$ cm²	1+1+1
17	Pond $= 231$ m²; remaining $= 1386 - 231 = 1155$ m²	2+1
	OR: $\frac{\theta}{360} \times 113.04 = 18.84 \Rightarrow \theta = 60^\circ$; arc $= \frac{1}{6} \times 37.68 = 6.28$ cm	2+1
18	$89.83 + 308 = 397.83$ cm³	1+1+1
	OR: $l = 5$: $62.8 + 100.48 = 163.28$ cm²	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{616}{8} = 77$ cm² (ii) $\frac{88}{8} = 11$ cm (iii) $616 - 231 = 385$ cm²	2+1+2
	OR: (i) $\frac{1}{4} \times 1386 = 346.5$ m² (ii) $1600 - 346.5 = 1253.5$ m² (iii) $616 - 346.5 = 269.5$ m²	2+1+2
20	(i) 753.6 cm³ (ii) 133.97 cm³ (iii) 887.57 cm³ ≈ 0.89 L (iv) 376.8 cm²	1+1+2+1
	OR: (i) $108\pi + 12\pi = 120\pi \approx 377.14$ cm³ (ii) $l = 5$: $72\pi + 9\pi + 15\pi = 96\pi \approx 301.71$ cm²	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 11 & 12: **Areas Related to Circles, Surface Areas and Volumes** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 154 cm²	$\frac{1}{4} \times 616$
2	**(a)** $\frac{22}{3}$ cm	$\frac{1}{6} \times 44$
3	**(d)** 14 cm	$r^2 = 196$
4	**(a)** Incorrect — 44 cm is the circumference; area is 154 cm²	$\pi r^2 = 154$
5	**(a)** Student A is correct	30° per hour
6	**(b)** 308 cm³	154×2
7	**(a)** 1232 cm²	2×616
8	**(b)** 1 : 3	$\frac{1}{3}\pi r^2 h : \pi r^2 h$
9	**(c)** Incorrect — it is a cylinder joined with a cone	Conical tip
10	**(a)** Student A is correct	Two faces hidden

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{1}{8} \times 1386 = 173.25$ cm²	2
12	$\frac{1}{10} \times 62.8 = 6.28$ cm	2
13	$384 - 16\pi + 32\pi = 384 + 16\pi \approx 434.29$ cm²	1+1
14	$90\pi + 12\pi = 102\pi \approx 320.57$ cm³	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$113.04 - 72 = 41.04$ cm²	1+1+1
16	$660 + 154$ (other base) $+308 = 1122$ cm²	1+1+1
17	Area 235.5 cm²; arc 31.4 cm; perimeter 61.4 cm	1+1+1
	OR: 240°: $\frac{2}{3} \times \frac{22}{7} \times 110.25 = 231$ cm²	1+1+1
18	$110 + 176 = 286$ m²	1+1+1
	OR: $\frac{2}{3}\pi(729) = 486\pi \approx 1527.43$ cm³ ≈ 1.53 L	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 22 cm (ii) 231 cm² (iii) $231 - 190.73 = 40.27$ cm² (iv) $22 + 21 = 43$ cm	1+1+2+1
	OR: (i) $2 \times 462 = 924$ cm² (ii) 462 cm² (iii) 18.48%	2+1+2
20	(i) 282.6 m³ (ii) 37.68 m³ (iii) 320.28 m³ (iv) $l = 5$: $188.4 + 47.1 = 235.5$ m²	1+1+1+2
	OR: (i) $231 + 89.83 = 320.83$ m³ (ii) $132 + 77 = 209$ m² (iii) ₹2090	2+2+1

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ANSWER KEY & MARKING SCHEME SET C

Chapters 11 & 12: **Areas Related to Circles, Surface Areas and Volumes** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 37.68 cm²	$\frac{1}{3} \times 113.04$
2	**(d)** 11 cm	$\frac{1}{8} \times 88$
3	**(c)** 60°	$\frac{22}{132} \times 360^\circ$
4	**(a)** Correct	Area $\propto \theta$
5	**(a)** Student A is correct	Half of 616
6	**(a)** 36π cm³	$\frac{4}{3}\pi \times 27$
7	**(b)** 60π cm²	$l = 10$
8	**(b)** 462 cm²	$3\pi r^2$
9	**(a)** Incorrect — a sphere has area $4\pi r^2$	Flat faces hidden
10	**(b)** Student A is correct	Hemisphere $= \frac{2}{3}\pi r^3$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{1}{5} \times 154 = 30.8$ cm²	2
12	$\frac{1}{10} \times 220 = 22$ cm	2
13	$718.67 + 770 = 1488.67$ cm³	1+1
14	Cuboid $6 \times 2 \times 2$: $2(12 + 4 + 12) = 56$ cm²; claim false (72 = three separate cubes)	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$86.625 - 55.125 = 31.5$ cm²	1+1+1
16	$504\pi - 144\pi = 360\pi \approx 1131.43$ cm³	1+1+1
17	108°; area $= 0.3 \times 154 = 46.2$ cm²	1+2
	OR: $19.625 - 12.5 = 7.125$ cm²	1+1+1
18	$90\pi + 2 \times 12\pi = 114\pi \approx 358.29$ cm³	1+1+1
	OR: $l = 7$: $92.4 + 110.88 = 203.28$ cm²	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{5}{12} \times 616 \approx 256.67$ m² (ii) $\frac{5}{12} \times 88 \approx 36.67$ m (iii) ≈ 359.33 m²	2+1+2
	OR: (i) $\frac{1}{9} \times \frac{22}{7} \times 147 \approx 51.33$ m² (ii) $9.78 + 4.89 + 7 + 7 \approx 28.67$ m	3+2
20	(i) $10\pi + \frac{2}{3}\pi = \frac{32}{3}\pi \approx 33.52$ cm³ (ii) $20\pi + 2\pi = 22\pi \approx 69.14$ cm² (iii) $1000 \div 33.52 \approx 29.8 \Rightarrow 29$	2+2+1
	OR: (i) $0.5\pi + \frac{\pi}{6} = \frac{2\pi}{3} \approx 2.10$ m³ (ii) $2\pi + \pi = 3\pi \approx 9.43$ m²	3+2

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ANSWER KEY & MARKING SCHEME SET D

Chapters 11 & 12: **Areas Related to Circles, Surface Areas and Volumes** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 346.5 cm²	$\frac{1}{4} \times 1386$
2	**(b)** 44 cm	$\frac{1}{4} \times 176$
3	**(c)** 308 cm²	Semicircle
4	**(a)** Correct	Half the circle
5	**(b)** Student A is correct	$22 + 14 + 14$
6	**(d)** 88 cm³	$\frac{22}{7} \times 4 \times 7$
7	**(b)** 24π cm²	$15\pi + 9\pi$
8	**(d)** 718.67 cm³	$\frac{2}{3} \times \frac{22}{7} \times 343$
9	**(a)** Incorrect — it is one-third	$\frac{1}{3}\pi r^2 h$
10	**(a)** Student A is correct	Volumes add

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{1}{4} \times 38.5 = 9.625$ cm²	2
12	$\pi r^2 = 154 \Rightarrow r = 7$ cm	2
13	$216 - 9\pi + 18\pi = 216 + 9\pi \approx 244.29$ cm²	1+1
14	$616 + 154 = 770$ cm³	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$18.84 - 15.57 = 3.27$ cm²	1+1+1
16	Volume $24\pi \approx 75.43$ cm³; surface $30\pi \approx 94.29$ cm²	1½+1½
17	35 min $= 210°$: $\frac{210}{360} \times 1386 = 808.5$ cm²	1+1+1
	OR: Arc 66, perimeter 94 cm; area 462 cm²	1+1+1
18	$44 + 55 = 99$ m²; ₹9900	2+1
	OR: $200 - 3 \times 4.62 = 186.14$ cm³	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 62.8 m² (ii) 12.56 m (iii) 5 (iv) $4 \times 62.8 = 251.2$ m²	1+1+1+2
	OR: (i) 51.33 (ii) $\frac{49\sqrt{3}}{4} \approx 21.19$ (iii) ≈ 30.14 (iv) ≈ 123.86 cm²	1+1+2+1
20	(i) $314 + 261.67 = 575.67$ cm³ (ii) $125.6 + 157 = 282.6$ cm² (iii) ₹565.20	2+2+1
	OR: (i) $1540 + 1232 + 718.67 = 3490.67$ cm³ (ii) $440 + 550 + 308 = 1298$ cm²	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.